Feature Engineering Master Guide

This document summarizes the purpose of every engineered feature, the variables used in each research question, and suggested analyses for the EDA and SQL phases. It is intended to serve as the team's reference during the remainder of the project.

1. Engineered Features Summary

Feature	Purpose	Supports
match_result	Classifies each match as Home Win, Away Win or Draw.	RQ1,2,3,4
goal_difference	Goal margin between home and away teams.	RQ1,RQ3,RQ4
total_goals	Total goals scored in a match.	RQ1
total_xg	Combined expected goals (xG).	RQ1
xg_difference	Difference in expected goals.	RQ3
points_difference	Difference in league points before the match.	RQ2
recent_form_difference	Difference in points earned over the previous 4 matches.	RQ2
average_home_team_rating	Average of defense, midfield and attack ratings.	RQ4
average_away_team_rating	Average of defense, midfield and attack ratings.	RQ4
rating_difference	Difference between both average team ratings.	RQ4

RQ1. Is Expected Goals (xG) a good predictor of actual goals?

Dataset Columns

home_score
away_score
home_xg
away_xg
total_goals
total_xg
goal_difference
match_result

Suggested Univariate Analysis

Distribution of total_goals
Distribution of total_xg
Distribution of goal_difference

Suggested Bivariate Analysis

total_xg vs total_goals
xg_difference vs goal_difference
match_result by total_xg

Suggested SQL Analysis

AVG(total_goals) by league
AVG(total_xg) by season
Correlation support using grouped aggregates

Suggested Visualizations

Histogram
Scatter plot
Boxplot by league

RQ2. Does recent team form influence match outcomes?

Dataset Columns

home_points
away_points
home_points_last_4_matches
away_points_last_4_matches
points_difference
recent_form_difference
match_result

Suggested Univariate Analysis

Distribution of recent_form_difference
Distribution of points_difference

Suggested Bivariate Analysis

recent_form_difference vs match_result
points_difference vs match_result

Suggested SQL Analysis

AVG(recent_form_difference) by league
Win rate by points_difference buckets
Average points_difference by season

Suggested Visualizations

Histogram
Boxplot
Bar chart

RQ3. Does expected performance before the match influence the final result?

Dataset Columns

home_xg
away_xg

xg_difference
goal_difference
match_result

Suggested Univariate Analysis

Distribution of xg_difference

Suggested Bivariate Analysis

xg_difference vs goal_difference
xg_difference vs match_result

Suggested SQL Analysis

AVG(xg_difference) by league
Average xg_difference for wins/draws/losses

Suggested Visualizations

Scatter plot
Boxplot
Bar chart

RQ4. How do team player ratings and team formations influence match performance?

Dataset Columns

home_formation
away_formation
home_defense_rating
home_midfield_rating
home_attack_rating
away_defense_rating
away_midfield_rating
away_attack_rating
average_home_team_rating
average_away_team_rating

rating_difference
match_result
goal_difference

Suggested Univariate Analysis

Distribution of rating_difference
Distribution of formations

Suggested Bivariate Analysis

rating_difference vs goal_difference
rating_difference vs match_result
formation vs match_result

Suggested SQL Analysis

AVG(rating_difference) by formation
Win percentage by formation
Average ratings by league

Suggested Visualizations

Histogram
Boxplot
Bar chart
Scatter plot